Problem Chosen	2026 MCM/ICM Summary Sheet	Team Control Number
C		1111111

Data With the Stars: Inferring Hidden Fan Support Under Voting-Rule Constraints

Summary

Dancing with the Stars combines expert evaluation with audience voting, but the fan-vote component is secret. This makes Problem C fundamentally different from a direct prediction problem: the observed eliminations do not reveal a unique set of fan votes. Our solution therefore treats **fan support as a latent variable**. Instead of pretending to know exact vote totals, we reconstruct weekly competitions, translate observed eliminations into feasible fan-support constraints, estimate maximum-entropy fan-share distributions, and use those estimates to compare voting rules and propose a producer-facing rule.

Using the official data set of 421 contestants across 34 seasons, we rebuilt **2,777 active contestant-week records** and modeled **261 elimination weeks**. The resulting latent-support model exactly reproduced the observed modeled eliminations by construction and confirmed that the underlying fan-vote uncertainty is large. This **non-identifiability** is the central practical finding: many fan-support profiles can explain the same observed elimination. It means that a good voting system should not merely chase an estimated "true" fan vote, but should be robust to close and ambiguous cases.

Our counterfactual simulation shows that rank and percent rules disagree in **26.4%** of modeled elimination weeks, while estimated fan support changes the judge-only bottom set in **63.6%** of modeled weeks. We also examine controversial contestants, professional-dancer and celebrity-category effects, and the institutional role of bottom-two judge saves. These results support a hybrid recommendation: use a **percent-based combined score** for transparency, but introduce a **gray-zone judge save** when the bottom-two combined margin is small. In our simulation, the gray-zone trigger activates in 61.7% of modeled elimination weeks, making close calls explicit rather than hiding them inside opaque vote totals.

The baseline tests make the institutional stakes clearer. A **judge-only rule** matches the observed eliminated set in only 36.4% of modeled weeks, while a rank-rule counterfactual using the same inferred fan support matches 73.6%. The largest fan-rescue gaps reach 15 rank positions, showing that viewer support can dramatically offset weak judge placement. We also find that many controversial outcomes are not random noise: they arise when expert rank and feasible audience rank point in opposite directions for several consecutive weeks.

For producers, the recommendation is simple: keep fan influence visible, avoid overclaiming exact fan support, and reserve judge discretion for statistically fragile bottom-two cases. The proposed system is more explainable than a pure rank system and more stable than treating a tiny combined-score gap as decisive. It also creates a public narrative that can be defended: ordinary weeks are decided by transparent percentages, while close calls are explicitly labeled as close calls and resolved by a limited judge-save procedure.

The paper's broader lesson is that hidden public votes should be modeled as constraints, not guessed labels. This preserves mathematical honesty, gives producers actionable diagnostics, and avoids recommending a voting system that looks precise only because it ignores the uncertainty in the fan-vote component.

Contents

1 Introduction

Dancing with the Stars (DWTS) is a useful voting-system case study because it combines two sources of evaluation that are deliberately different: judges score dance quality, while the public votes for contestants. The official Problem C data set reports contestant information, final outcomes, placements, and weekly judges' scores, but it does not reveal the fan votes. The hidden vote component creates the central modeling difficulty: a contestant can be eliminated under many different fan-vote profiles, and the same public outcome does not identify a unique distribution of fan support.

This observation changes the problem from a direct prediction task into an **inverse problem**. A supervised model that "predicts" fan votes would be unverifiable because no fan-vote labels exist in the data. Instead, we use the known voting systems as constraints. For each week, the observed elimination is treated as evidence that the true fan support must have belonged to a feasible set. We then estimate a conservative point inside that set and use the uncertainty itself as part of the answer.

Our approach has **four linked layers**. First, we reconstruct each weekly competition from the wide official data. Second, we formalize the rank, percent, and bottom-two judge-save mechanisms. Third, we estimate latent fan support using feasible-set constraints and a maximum-entropy principle. Fourth, we use the estimated support profiles for counterfactual simulations and a producer-facing voting recommendation. Figure 1 summarizes the complete workflow.

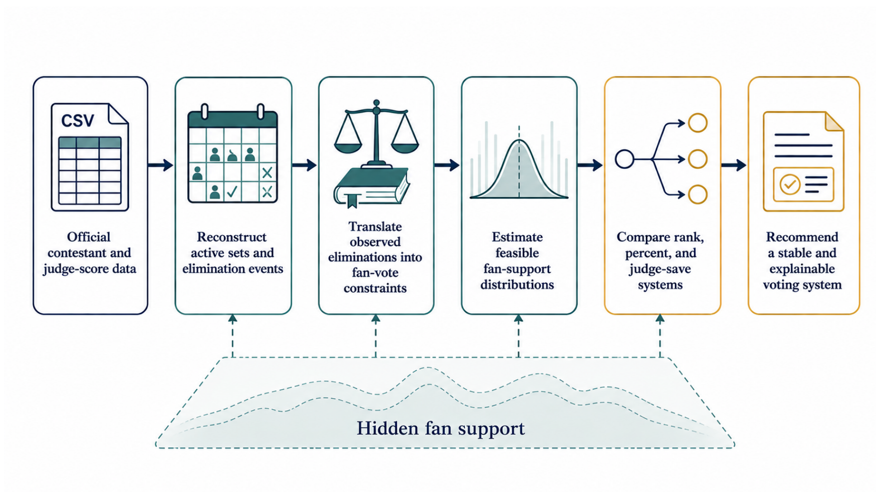


Figure 1: Overall workflow for hidden fan-support inference.

The main contribution of the paper is not a claim that exact fan votes can be recovered. Rather, we show how to quantify what can be inferred from the publicly observable record, how different voting rules behave under the same latent support estimates, and how producers can design a rule that remains transparent when the bottom of the leaderboard is statistically fragile.

1.1 Problem restatement

The official problem asks for a data-driven analysis of the DWTS voting system. We restate the modeling tasks as four operational questions.

1. Reconstruct each season-week competition from the official contestant data and identify the active set, judge scores, and observed eliminations.
2. Infer what can be learned about hidden fan support without assuming unobserved vote totals.
3. Compare plausible voting rules under the same judge scores and latent fan-support estimates,

especially rank aggregation, percent aggregation, and bottom-two judge saves.

4. Recommend a transparent producer-facing voting system that balances expert judgment, viewer influence, and robustness in close elimination weeks.

These questions deliberately separate **historical reconstruction** from **policy recommendation**. A model that exactly reproduces past eliminations is useful as a consistency check, but the design decision for producers depends on how rules behave when the bottom-two margin is small and hidden fan support is non-identifiable.

1.2 Evaluation criteria

Because the problem is about voting-system design rather than only historical prediction, we evaluate candidate rules with four criteria. These criteria also guide the recommendation section, where mathematical performance must be translated into a rule that producers can explain on air.

Table 1: Voting-system design criteria used throughout the paper.

Criterion	Mathematical meaning	Producer-facing meaning
Reconstructability	The rule can be simulated from official judge scores and a feasible fan-support set.	The outcome can be audited after the show.
Transparency	Judge and fan components are measured on comparable scales.	Viewers can understand why a contestant is at risk.
Fan influence	The public component can change the judge-only bottom set.	Voting feels meaningful rather than symbolic.
Close-call stability	Tiny bottom-two gaps are not treated as certain.	Controversial eliminations have a defensible procedure.

2 Notation and Assumptions

Table 2: Core notation used in the modeling workflow.

Symbol	Type	Meaning
s	index	DWTS season.
t	index	Competition week within a season.
i	index	Active contestant in a season-week.
$A_{s,t}$	set	Contestants active in season s, week t.
$E_{s,t}$	set	Contestants eliminated after week t.
J_i	scalar	Total judge score for contestant i.
P_i^J	scalar	Judge-score share, $J_i/\sum_k J_k$.
F_i	scalar	Hidden fan-share variable.
C_i^{pct}	scalar	Percent-rule combined score.
C_i^{rank}	scalar	Rank-rule combined score.
τ	scalar	Gray-zone threshold for bottom-two margins.

Assumption 1 (Official-data sufficiency)**.** *The official CSV is treated as the authoritative source for contestant identity, weekly judge scores, placements, and final results. We do not add unverified external popularity data because fan votes are not observed in the official record.*

Assumption 2 (Active-set reconstruction)**.** *A contestant is active in a week if at least one valid judge score is present and the weekly judge total is positive. Once a contestant disappears from the active set, the disappearance is interpreted as an elimination unless the whole season has ended.*

Assumption 3 (Rule-consistency inference). *Observed eliminations are assumed to be consistent with the reported judge scores and a hidden fan-support vector under a percent-rule feasible set. This assumption is not a claim that DWTS always used an identical rule; it is the common constraint system used to compare historical outcomes and candidate rules on a shared scale.*

Assumption 4 (No causal claim from observational effects). *Professional-dancer, industry, age, and survival models are interpreted as association models. They can identify recurring patterns in the data, but they cannot prove that a dancer, age, or celebrity category caused a specific elimination outcome.*

3 Data Audit and Preprocessing

3.1 Raw data audit

The official CSV contains **421 contestant records**, 53 columns, and 34 seasons. For each contestant, the data include identity variables, celebrity industry, home location, age during season, final result, placement, and four possible judge-score columns for each week from Week 1 to Week 11. The raw file is a wide table: one row is one contestant, while weekly scores appear in repeated columns. This layout is convenient for storage but not for modeling a weekly elimination system.

We therefore first audited the data before fitting any model. Missing score cells can have several meanings: a fourth judge may not exist in a given week, a season may not include that week, or a contestant may already have been eliminated. In contrast, explicit zeros after a contestant leaves the competition should not be interpreted as poor performance by an active contestant. This distinction is important because the fan-vote constraints are constructed week by week from the active set.

Table 3: Data-processing decisions used before modeling.

Issue	Treatment	Reason
Wide weekly score columns	Convert to season-week-contestant panel.	Voting rules operate on weekly active sets, not full-season rows.
Missing fourth-judge cells	Use available judge scores only.	Some weeks have fewer judges; treating missing cells as zero would bias scores.
Post-elimination zeros	Exclude from active set.	A zero after exit is absence from the contest, not a low performance score.
Multi-elimination weeks	Keep all eliminated contestants in $E_{s,t}$.	These weeks impose multiple eliminated-survivor inequalities.
No-elimination weeks	Use for audit, not fan inference.	They do not identify a bottom set.

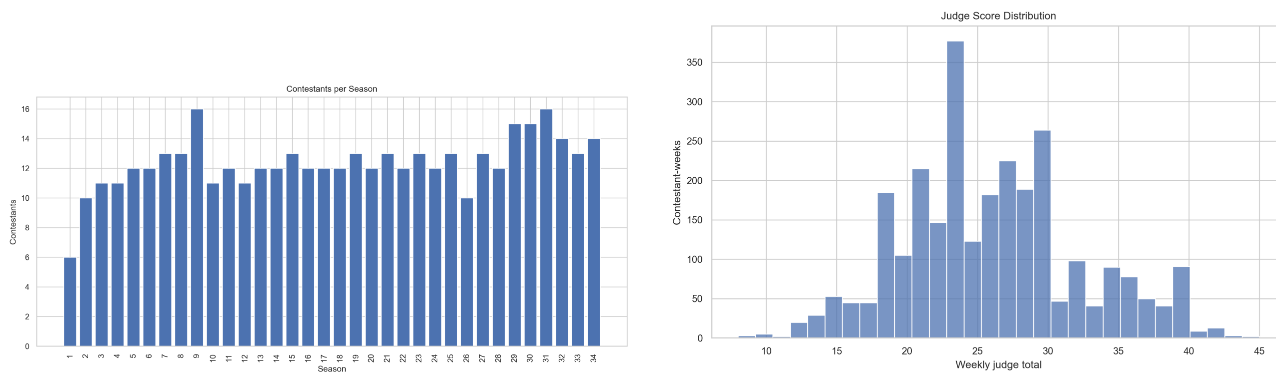
Table 4: Core data audit metrics used by the modeling workflow.

Metric	Value
Contestant records	421
Raw columns	53
Seasons	34
Reconstructed panel rows	4,631
Active contestant-week rows	2,777
Reconstructed elimination-event rows	343
Single-elimination rows	221
Multi-elimination rows	82
No-elimination rows	40

3.2 Panel reconstruction

We converted the wide table into a **long panel** indexed by season, week, and contestant. For each contestant-week we computed the available judge count, judge total, judge mean, and an active-status flag. A contestant is active in a week if at least one judge score is present and the weekly judge total is positive. This rule avoids treating post-elimination zeros as active competition records.

Next, we reconstructed the weekly active sets and observed elimination events. Let $A_{s,t}$ be the active set in season s and week t. A contestant in $A_{s,t} \setminus A_{s,t+1}$ is recorded as eliminated after week t. If this set contains multiple contestants, the week is marked as a multi-elimination event. If the active set does not shrink, the week is marked as a no-elimination week. This reconstruction produced 2,777 active contestant-week rows and 261 modeled elimination weeks used by the fan-support inference model.



(a) Contestants per season. **(b)** Active weekly judge totals.

Figure 2: Data reconstruction diagnostics.

3.3 Reconstruction diagnostics

The reconstructed active-set matrix is a useful diagnostic because many later calculations depend on having the right contestants in the right week. A season with a missing or duplicated active block would produce visible irregularities in Figure 3. The heatmap shows the expected staircase pattern: seasons begin with larger casts and contract over time, while a small number of weeks remain flat because no elimination occurs.

This diagnostic matters because fan-support inference is local to a season-week. A contestant excluded from $A_{s,t}$ would not receive a fan-share variable, while a contestant incorrectly kept active after elimination would distort the simplex constraint $\sum_i F_i = 1$. We therefore treat active-set reconstruction as a model input rather than a bookkeeping detail.

4 Voting Rule Formalization

The contest history involves several ways of combining judges and viewers. We formalize three mechanisms: a rank rule, a percent rule, and a bottom-two judge-save rule. Figure 4 gives the conceptual comparison.

For season s, week t, and active contestant $i \in A_{s,t}$, let J_i be the observed total judge score and let F_i be the unknown fan-share variable. We normalize fan shares so that

$$F_i \geq 0, \qquad \sum_{i \in A_{s,t}} F_i = 1.$$

Under a **percent rule**, judge scores and fan shares are both expressed as shares. The normalized

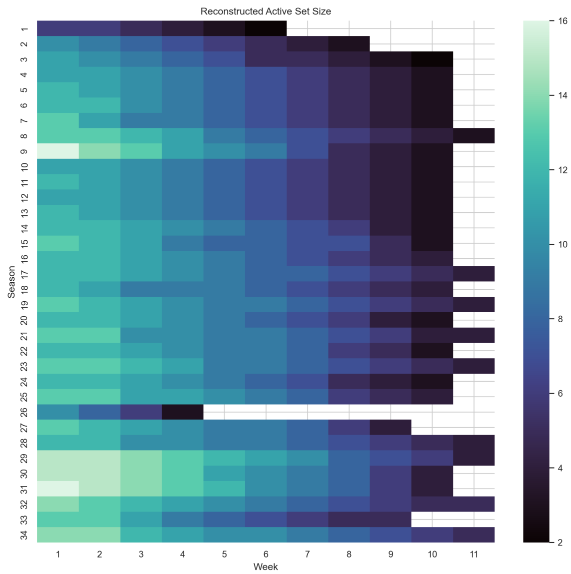


Figure 3: Reconstructed active contestants by season and week.

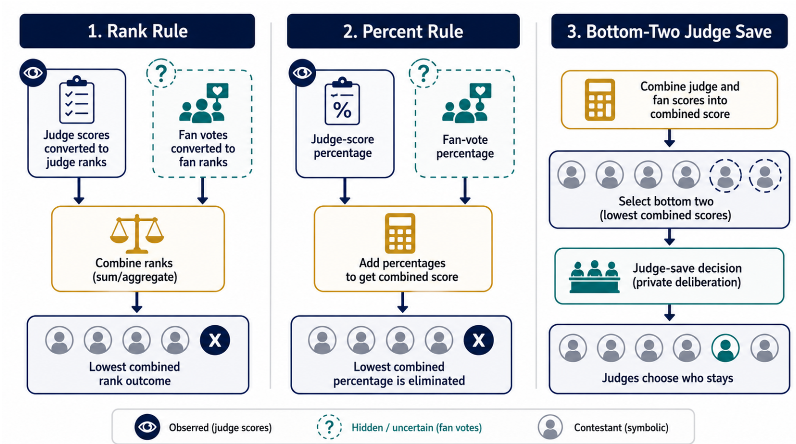


Figure 4: Rank, percent, and gray-zone voting mechanisms.

judge share is

$$P_i^J = \frac{J_i}{\sum_{k \in A_{s,t}} J_k},$$

and the combined score is

$$C_i^{\text{pct}} = P_i^J + F_i.$$

The lowest combined score is eliminated. If the observed eliminated set for a week is $E_{s,t}$ and the survivors are $S_{s,t} = A_{s,t} \setminus E_{s,t}$, then every eliminated contestant must have combined score no larger than every survivor:

$$P_e^J + F_e \leq P_j^J + F_j, \qquad e \in E_{s,t},\ j \in S_{s,t}.$$

These inequalities define a feasible fan-support polytope. Equivalently, for each eliminated-survivor pair $m = (e, j)$,

$$F_e - F_j \leq P_j^J - P_e^J.$$

Stacking all such pairwise constraints gives

$$AF \leq b, \qquad \mathbf{1}^\top F = 1, \qquad F \geq 0,$$

where one row of A has a $+1$ in the eliminated contestant's column and a -1 in the survivor's column. Thus the inverse voting problem is a convex feasibility problem. For a week with n active contestants and k eliminations, the number of pairwise rule constraints is

$$k(n-k).$$

This is why multi-elimination weeks carry more information than a single elimination with the same active set.

Under a **rank rule**, the judge score and fan support are converted into ranks:

$$C_i^{\text{rank}} = R_i^J + R_i^F,$$

where larger C_i^{rank} means a worse combined rank. The contestant or contestants with the worst combined rank are eliminated. We use the maximum-entropy fan-share estimate to induce fan ranks for rank-rule counterfactuals.

The **bottom-two judge-save** mechanism is represented as a two-stage process. First, a combined score identifies the bottom two contestants. Second, judges choose which contestant to save. Because detailed judge-save votes are not fully observed in the data, we use it as a policy simulation rather than a direct historical reconstruction.

Table 5: Voting-rule formalization used in the simulations.

Rule	Score	Elimination decision
Percent	judge share + fan share	lowest combined percent
Rank	judge rank + fan rank	worst combined rank
Bottom-two judge save	combined score, then judge decision	judge save among bottom two

5 Fan Vote Estimation

5.1 Why fan votes are not directly predictable

The fan vote is hidden and **non-identifiable** from the observed data alone. If a contestant is eliminated, we learn that the combined judge-fan result placed that contestant at or near the bottom under the relevant rule. We do not learn the exact vote total, the absolute number of votes, or the unique share of each contestant. This is why our model estimates feasible support rather than exact votes.

5.2 Feasible fan-support intervals

For each modeled elimination week, we build the percent-rule linear constraints from the observed eliminated set. We then solve two linear programs for each contestant i:

$$\underline{F}_i = \min_F \{F_i : AF \le b,\ \mathbf{1}^\top F = 1,\ F \ge 0\}, \qquad \overline{F}_i = \max_F \{F_i : AF \le b,\ \mathbf{1}^\top F = 1,\ F \ge 0\}$$

The result is an interval $[\underline{F}_i, \overline{F}_i]$ that describes how much fan support contestant i could have had while still being consistent with the observed elimination. We used SciPy's linear programming solver for this step [6]. Across the 261 modeled weeks, the workflow solves 2,252 pairwise elimination constraints, with a maximum of 30 constraints in a single week.

Table 6: Constraint-system diagnostics for the fan-support inverse problem.

Diagnostic	Value	Unit
Modeled elimination weeks	261	season-week events
Mean active contestants	8.62	contestants
Median active contestants	9.00	contestants
Total linear inequalities	2,252	constraints
Mean inequalities per week	8.63	constraints
Maximum inequalities in a week	30	constraints
Median bottom-two margin	7.53×10^{-6}	combined share

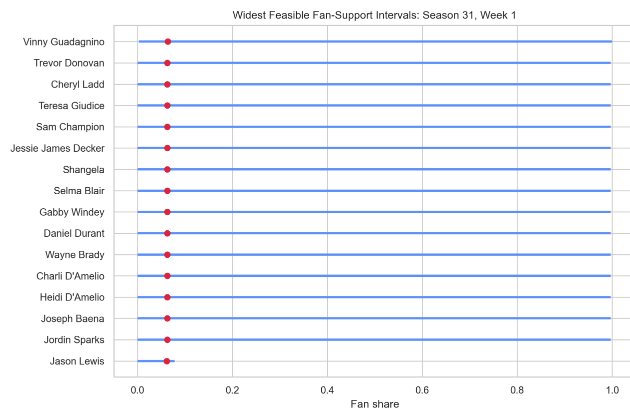


Figure 5: Feasible fan-support intervals in a high-uncertainty week.

5.3 Maximum-entropy point estimate

Intervals are useful for uncertainty, but **counterfactual simulations** also need a single support profile. We choose the maximum-entropy feasible profile:

$$\max_{F} \; -\sum_{i \in A_{s,t}} F_i \log F_i$$

subject to the feasible-set constraints. The entropy objective selects the least concentrated fan-support vector consistent with the observed outcome. It therefore avoids arbitrarily inventing a dramatic fan favorite unless the elimination constraints require one. This follows the maximum-entropy principle introduced by Shannon [3] and is implemented as a convex optimization problem using CVXPY [5].

The Lagrangian form also shows why the estimate is conservative. With nonnegative multipliers μ for $AF \le b$ and scalar λ for $\mathbf{1}^\top F = 1$, the stationarity condition is

$$-\log F_i - 1 - \lambda - \sum_m \mu_m A_{m,i} = 0,$$

so any interior solution has

$$F_i = \exp\left(-1 - \lambda - \sum_m \mu_m A_{m,i}\right).$$

Only active elimination constraints tilt the profile away from a nearly uniform distribution. When the observed outcome imposes weak restrictions, the maximum-entropy solution stays diffuse rather than inventing unsupported public-vote spikes.

Table 7: Fan-support inference algorithm used for each modeled elimination week.

Step	Operation
1	Build the active set $A_{s,t}$ from reconstructed contestant-week records.
2	Normalize judge totals into judge shares P_i^J.
3	Translate each observed eliminated-survivor pair into a linear inequality $P_e^J + F_e \le P_j^J + F_j$.
4	Solve linear programs for each contestant's feasible lower and upper fan-share bounds.
5	Solve the maximum-entropy convex program inside the feasible polytope.
6	Use the resulting support vector for validation, rule counterfactuals, controversial-case diagnostics, and the proposed-system simulation.

The algorithm produces three different outputs from the same constraints: intervals, a point estimate, and diagnostic flags. This is important because a single estimated fan-share vector should not be treated as ground truth. The intervals tell us how much freedom remains after the historical elimination is enforced, while the maximum-entropy vector gives a stable reference point for comparing rules.

5.4 Identifiability boundary

The hidden-vote problem has an important boundary: the data can identify a feasible region, but not a unique public-vote vector. For a week with n active contestants, the fan simplex has dimension $n-1$. If r independent elimination inequalities are active at the solution, the locally identified face has dimension approximately

$$d_{\text{face}} = (n-1) - r.$$

Most weeks have fewer active constraints than contestants, so d_{face} is positive. This explains why a perfect historical reconstruction does not mean the exact fan vote has been recovered. It means the observed elimination is compatible with at least one point inside a larger feasible polytope.

We therefore report both point estimates and uncertainty widths. The point estimate answers "what is the least concentrated support profile consistent with the outcome?" The interval width answers a different question: "how much room does the official record leave for alternative support profiles?" A contestant with a wide interval should not be described as precisely popular or unpopular. The responsible conclusion is that the public evidence is weak for that contestant-week.

Table 8: Interpretation of fan-support outputs.

Output	Mathematical role	Interpretation
Feasible interval	Projection of the polytope onto F_i.	Range of fan shares consistent with the elimination.
Entropy estimate	Interior point maximizing $-\sum_i F_i \log F_i$.	Conservative reference profile for simulations.
Interval width	$\overline{F}_i - \underline{F}_i$.	Degree of non-identifiability for contestant i.
Bottom-two margin	Gap between the two lowest combined scores.	Practical fragility of the elimination decision.

5.5 Uncertainty as a model output

The average feasible interval width across modeled weeks is 0.862, which is large on the $[0,1]$ fan-share scale. This does not mean the model is weak; it means the historical outcomes do not uniquely identify fan votes. That uncertainty is itself a substantive result. It warns against overinterpreting any one estimated fan-support profile and motivates a voting rule that handles close bottom-two cases explicitly.

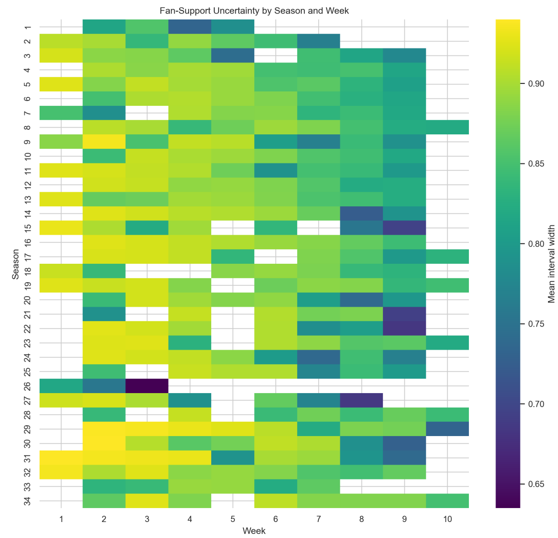


Figure 6: Fan-support uncertainty by season and week.

6 Model Validation and Baseline Tests

The validation step asks a limited but essential question: after reconstructing the active set and solving for a feasible maximum-entropy support vector, does the model reproduce the observed elimination under the percent-rule constraints? The answer is yes for all 261 modeled elimination weeks. This 100% reproduction rate should not be misread as out-of-sample prediction. It is a consistency check showing that the reconstructed data, constraints, and optimization pipeline agree with the observed historical record.

Table 9: Validation and uncertainty metrics.

Metric	Value
Modeled elimination weeks	261
Percent-rule reproduction rate	100.0%
Observed elimination in modeled bottom two	100.0%
Mean feasible fan-share interval width	0.862

The more important validation result is the scale of uncertainty. The average interval width of 0.862 shows that observed eliminations alone leave the fan support profile highly underdetermined. In practical terms, a reported elimination may be consistent with a wide range of public-vote configurations. This strengthens the case for comparing voting systems through robust counterfactual behavior rather than through a single supposed true vote count.

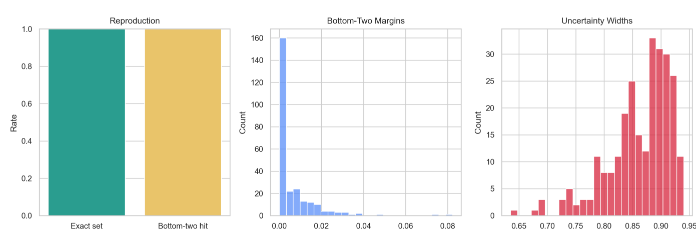


Figure 7: Validation dashboard for reconstruction and uncertainty.

6.1 Baseline comparison

The **consistency check** above is necessary but not sufficient. Because the percent-rule fan vector is constructed to satisfy the observed elimination constraints, exact reproduction should be interpreted as feasibility rather than predictive accuracy. We therefore compare it with three simpler baselines: a rank-rule counterfactual using the same inferred fan support, a judge-only bottom set, and a fan-only bottom set.

Table 10: Baseline and counterfactual rule comparison over 261 modeled weeks.

Candidate rule	Exact match	Mean overlap	Interpretation
Percent + inferred fan	1.000	1.000	constraint reconstruction
Rank + inferred fan	0.736	0.760	rule-change counterfactual
Fan-only bottom	0.625	0.681	audience-only baseline
Judge-only bottom	0.364	0.416	no-fan baseline

The judge-only baseline matches the historical eliminated set in only 36.4% of modeled weeks, while the fan-only bottom set matches 62.5%. This does not mean that fan preference alone determined eliminations; judge scores still shape the feasible set and the combined score. It does show that hidden fan support is not a small perturbation around the judge leaderboard. The rank-rule counterfactual matches 73.6% of historical eliminated sets, leaving more than one quarter of weeks where the aggregation rule itself matters.

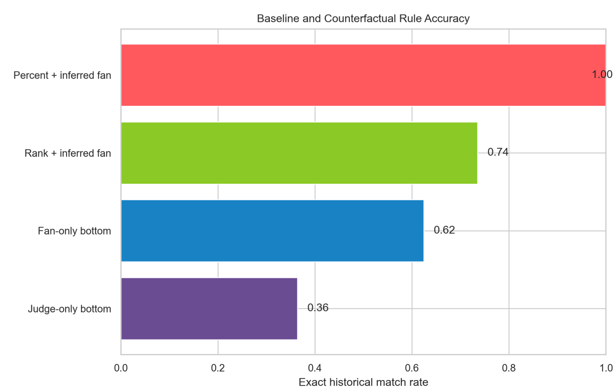


Figure 8: Baseline and counterfactual exact-match rates.

For a candidate rule r, we compute exact-set match and observed-set overlap as

$$M_r = \frac{1}{T}\sum_{t=1}^{T} \mathbf{1}\{\widehat{E}_{r,t} = E_t\}, \qquad O_r = \frac{1}{T}\sum_{t=1}^{T} \frac{|\widehat{E}_{r,t} \cap E_t|}{|E_t|}.$$

The overlap score is useful for multi-elimination weeks, where a rule can get one eliminated contestant right but not the full eliminated set.

Table 11: Rule diagnostics by elimination-event type.

Event type	Weeks	Mean active	Constraints	Rule diff.	Fan override
Single elimination	221	8.50	7.50	25.3%	60.2%
Multi elimination	40	9.33	14.88	32.5%	82.5%

Multi-elimination weeks are more informative because each eliminated contestant is compared with every survivor. They also show stronger fan override behavior, suggesting that group-elimination weeks are particularly sensitive to the hidden audience component.

Table 12: Rule diagnostics by season era.

Era	Weeks	Mean active	Rule diff.	Fan override	Mean width
S1–S10	78	7.74	26.9%	57.7%	0.858
S11–S20	81	8.37	22.2%	61.7%	0.869
S21–S30	70	9.21	34.3%	67.1%	0.853
S31–S34	32	10.12	18.8%	75.0%	0.877

7 Rank vs. Percent Counterfactual Analysis

We next use the same maximum-entropy fan-support estimates to compare the rank and percent rules. This isolates the effect of the rule itself: the contestants, judge scores, and estimated fan support remain fixed, while only the rule used to combine them changes.

Across 261 modeled elimination weeks, rank and percent rules produce different elimination sets in 26.4% of weeks. Estimated fan support changes the judge-only bottom set in 63.6% of weeks. These two metrics have different interpretations. The first measures institutional sensitivity to rule choice. The second measures how often the public component is strong enough to change what would have happened under judges alone.

Table 13: Counterfactual rule-comparison metrics.

Metric	Value
Modeled elimination weeks	261
Rank-percent outcome difference rate	26.4%
Fan override rate relative to judge-only bottom set	63.6%
Mean rank-percent eliminated-set overlap	0.893

The heatmap in Figure 9 shows that rule disagreement is not confined to a single season. It appears throughout the history of the data, which means that voting-rule design is not a minor technicality. A rule change can alter which contestants are at risk even when the same hidden support estimate is used.

The mechanism is simple. Under a percent rule, contestant i beats contestant j when

$$(P_i^J - P_j^J) + (F_i - F_j) > 0.$$

Under a rank rule, the same comparison is based on ordinal positions,

$$(R_j^J - R_i^J) + (R_j^F - R_i^F) > 0,$$

where better ranks have smaller numeric values. These two inequalities can have different signs because rank conversion replaces score gaps with unit steps. A one-point judge-score difference and a ten-point judge-score difference can both become adjacent rank positions. This is why the rule switch is most consequential in crowded early weeks: many contestants create many adjacent ranks, and small perturbations can move several names across the bottom boundary.

7.1 High-leverage rule-switch weeks

The most informative counterfactuals are not the average weeks but the weeks where the rule changes the at-risk contestants. Table 15 lists representative cases with large uncertainty and no overlap between the rank-rule and percent-rule eliminated sets. Season 9, Week 1 is especially diagnostic because it is a multi-elimination week: the percent rule reproduces Ashley Hamiliton and Macy Gray as the eliminated pair, while the rank rule would instead place Chuck Liddell and Debi Mazar at the bottom.

Table 14: Mechanisms that produce rank-versus-percent disagreement.

Mechanism	What happens mathematically	Practical effect
Score compression	Percent gaps are replaced by rank steps.	Large judge-score advantages can be understated.
Crowded leaderboard	Many contestants have adjacent ranks.	One fan-rank change can alter several bottom candidates.
Multi-elimination	More than one contestant crosses the elimination boundary.	Rule differences become easier for viewers to notice.
Hidden fan uncertainty	Many feasible fan profiles satisfy the observed outcome.	A deterministic rank result can look more precise than the data justify.

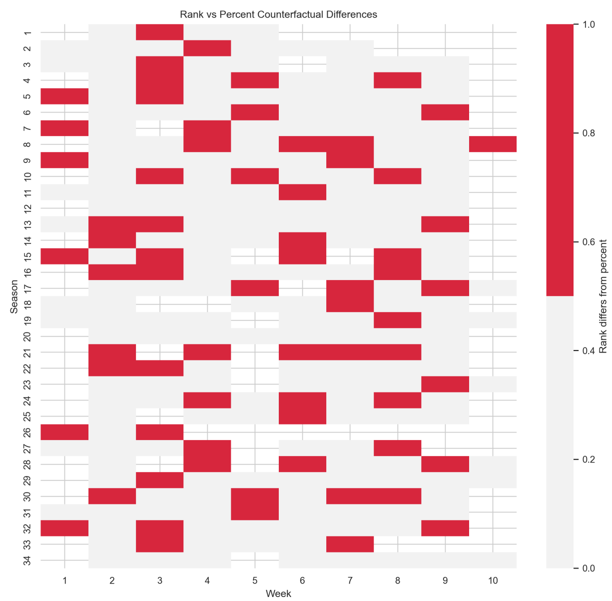


Figure 9: Season-week map of rank-versus-percent differences.

Table 15: Representative rank-versus-percent switch cases.

Season	Week	Observed	Percent rule	Rank rule	Mean width
9	1	Ashley Hamilton; Macy Gray	Macy Gray; Ashley Hamiliton	Chuck Liddell; Debi Mazar	0.885
30	2	Martin Kove	Martin Kove	Matt James	0.940
29	3	Carole Baskin	Carole Baskin	Justina Machado	0.936
32	1	Matt Walsh	Matt Walsh	Alyson Hannigan	0.935
15	1	Pamela Anderson	Pamela Anderson	Kirstie Alley	0.930

These cases explain why percent aggregation is preferable for transparency. In a rank system, small score differences can be converted into large rank jumps, especially early in a season when many contestants are active. Percent scoring keeps the magnitude of judge-score differences visible.

7.2 Fan-rescue diagnostics

We define a **fan-rescue gap** as

$$G_{i,t} = R^J_{i,t} - R^F_{i,t}.$$

A large positive value means that the contestant ranked poorly with judges but well with the inferred audience profile. The largest estimated rescue case is Vinny Guadagnino in Season 31, Week 1, whose judge rank was 16 while his maximum-entropy fan rank was 1.

Table 16: Largest estimated fan-rescue contestant-weeks.

Season	Week	Contestant	Judge rank	Fan rank	Gap	Result
31	1	Vinny Guadagnino	16	1	15	Eliminated Week 8
31	2	Cheryl Ladd	15	1	14	Eliminated Week 3
30	3	Cody Rigsby	14	1	13	3rd Place
31	3	Vinny Guadagnino	14	1	13	Eliminated Week 8
9	1	Michael Irvin	14	1	13	Eliminated Week 7

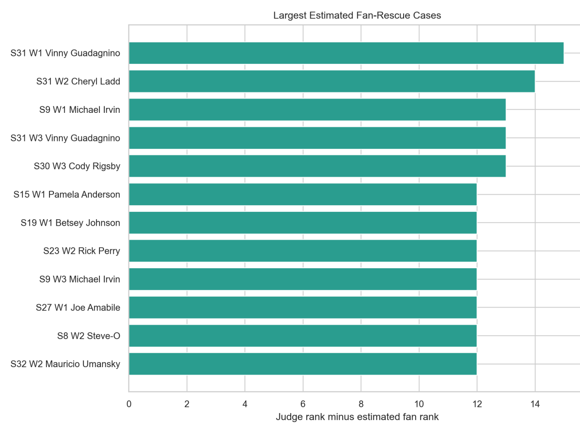


Figure 10: Largest estimated fan-rescue gaps.

7.3 Case interpretation

The case tables are not decorative; they identify the weeks where a rule change would be visible to viewers. Season 9, Week 1 is the cleanest rank-versus-percent example. The percent reconstruction puts Ashley Hamilton and Macy Gray at the bottom, matching the observed multi-elimination. The rank counterfactual instead moves Chuck Liddell and Debi Mazar into the eliminated set. Since the same judge scores and inferred fan profile are used, the difference comes from rank aggregation compressing score magnitudes into ordinal positions.

The fan-rescue table shows the opposite phenomenon: audience support can protect contestants who are weak on the judge leaderboard. Vinny Guadagnino's Season 31 trajectory is a strong example. In Week 1 he is last by judge rank but first by the maximum-entropy fan estimate, a 15-rank gap; he remains active until Week 8. Cody Rigsby's Season 30 run is also revealing because a large rescue gap occurs early, yet the contestant ultimately reaches 3rd Place. These cases support a rule design in which the fan vote is not merely a tie-breaker.

Finally, Joanna Krupa's Season 9 pattern shows that controversy can run in the other direction. Strong judge ranks combined with weaker estimated fan support create a risk of elimination that judges alone would not predict. A producer who wants to defend the show needs both types of diagnostics: fan-rescue cases and judge-favored but audience-weaker cases.

Table 17 summarizes how these cases should be read. The important point is that a controversial outcome is not automatically an unfair outcome. It becomes unfair-looking when the rule cannot explain whether the audience, judges, or a close combined margin drove the result. Our diagnostics separate those mechanisms before recommending a new rule.

Table 17: Diagnostic reading of representative high-leverage cases.

Case type	Model evidence	Rule-design implication
Fan rescue	Low judge rank but high estimated fan rank, as in Vinny Guadagnino's early Season 31 weeks.	The fan vote must remain powerful enough to overturn the judge leaderboard.
Judge-favored risk	High judge rank but weak estimated fan rank, as in Joanna Krupa's Season 9 pattern.	Eliminations can be defensible even when judges strongly prefer the contestant.
Persistent controversy	Multi-week judge-fan disagreement, as in Bobby Bones and Bristol Palin.	Producers need an audit trail, not a one-week explanation after criticism appears.
Rule-switch exposure	Different eliminated sets under rank and percent aggregation.	The official rule should use a score scale that viewers can interpret directly.

8 Controversial Contestants

The problem statement highlights cases where public popularity and judge scores may appear to pull in different directions. We therefore tracked Jerry Rice, Billy Ray Cyrus, Bristol Palin, and Bobby Bones using a controversy index,

$$CI_{i,t} = |R^J_{i,t} - R^F_{i,t}|,$$

where R^J is judge rank and R^F is the estimated fan rank. Large values indicate a large gap between expert evaluation and inferred public support.

Table 18: Summary of selected controversial contestants.

Contestant	Season	Weeks	Mean judge rank	Mean fan rank	Result
Bristol Palin	15	4	10.50	4.25	Eliminated Week 4
Bobby Bones	27	7	7.29	4.86	1st Place
Bristol Palin	11	9	6.22	3.89	3rd Place
Billy Ray Cyrus	4	7	5.86	4.00	Eliminated Week 8
Jerry Rice	2	7	4.86	4.14	2nd Place

The pattern is clearest for Bristol Palin's Season 15 run and Bobby Bones's Season 27 win. In both cases, estimated fan ranks are substantially better than judge ranks over multiple weeks. This does not prove a specific vote total, but it does quantify the direction of the conflict: these contestants survived longer than judge rank alone would suggest because the feasible fan-support estimates consistently place them higher with viewers.

8.1 Data-driven controversy scan

The four highlighted names are useful because they are recognizable examples, but the model can also scan all contestants for judge-fan instability. Table 19 ranks contestants by the maximum weekly gap between judge rank and estimated fan rank, requiring at least three modeled weeks so that a single anomalous week does not dominate the list.

This wider scan reveals two different kinds of controversy. Vinny Guadagnino, Michael Irvin, and Steve-O are fan-rescue cases: judges placed them relatively low, but the inferred audience ranks are much stronger. Joanna Krupa is the opposite pattern: strong judge ranks combined with weaker inferred fan support. Both patterns are important to producers because viewer complaints can arise from either side of the judge-audience divide.

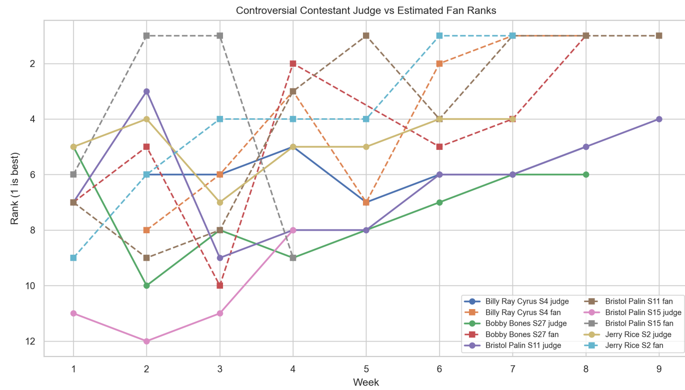


Figure 11: Judge rank versus estimated fan rank.

Table 19: Contestants with the largest judge-fan instability.

Contestant	Season	Weeks	Mean judge rank	Mean fan rank	Result
Vinny Guadagnino	31	8	10.88	4.50	Eliminated Week 8
Joanna Krupa	9	9	2.67	6.78	Eliminated Week 9
Cheryl Ladd	31	3	12.67	5.67	Eliminated Week 3
Michael Irvin	9	7	9.86	1.71	Eliminated Week 7
Cody Rigsby	30	8	8.12	4.25	3rd Place
Steve-O	8	5	10.20	2.00	Eliminated Week 6

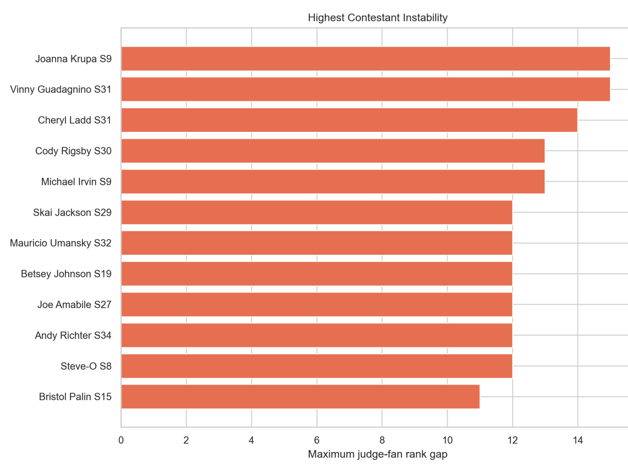


Figure 12: Contestant-level instability scan.

9 Professional Dancer and Celebrity Effects

We modeled three related effects: judge-score effects, fan-support effects, and survival-time effects. Judge and fan effects were estimated with ordinary least squares using celebrity age and industry indicators. Survival effects were estimated with a penalized Cox model using active-week duration and an elimination indicator. These models are exploratory rather than causal: the data are observational and contestants are not randomly assigned to industries or professional partners.

For contestant i in season s and week t, the score and fan-support models are

$$J_{i,s,t} = \beta_0 + \beta_1 \mathrm{Age}_i + \boldsymbol{\beta}_2^\top \mathrm{Industry}_i + \alpha_s + \varepsilon_{i,s,t},$$

$$\widehat{F}_{i,s,t} = \gamma_0 + \gamma_1 \mathrm{Age}_i + \boldsymbol{\gamma}_2^\top \mathrm{Industry}_i + \alpha_s + u_{i,s,t}.$$

The season term α_s absorbs broad era-level differences in casting, audience size, and scoring style. For survival, we use the proportional-hazard form

$$h_i(t \mid x_i) = h_0(t) \exp(\boldsymbol{\theta}^\top x_i),$$

where x_i contains the same contestant attributes. This common covariate structure makes the three effect estimates comparable even though they answer different questions.

The strongest consistent result is that age is associated with weaker estimated fan support and shorter survival. In the fan-support model, celebrity age has a negative coefficient of -0.00051 per year. In the survival model, age has a positive hazard coefficient of 0.0327, meaning older contestants tend to face earlier elimination after controlling for the included industry indicators. Some industry coefficients are also notable: for example, radio personalities and politicians receive lower judge-score estimates in the judge-score model, while social media personalities and TV personalities have lower estimated fan-share coefficients in the fan-support model. These findings should be interpreted as association patterns, not as causal claims.

Table 20: Selected exploratory effect estimates.

Effect	Estimate	Low	High	p-value
Judge score: age	-0.0455	-0.0497	-0.0413	1.5×10^{-91}
Fan support: age	-0.00051	-0.00066	-0.00036	3.0×10^{-11}
Survival hazard: age	0.0327	0.0244	0.0409	7.0×10^{-15}
Judge score: athlete	-0.333	-0.472	-0.195	2.5×10^{-6}
Judge score: model	-0.868	-1.184	-0.552	7.6×10^{-8}
Fan support: TV personality	-0.0096	-0.0149	-0.0043	3.7×10^{-4}

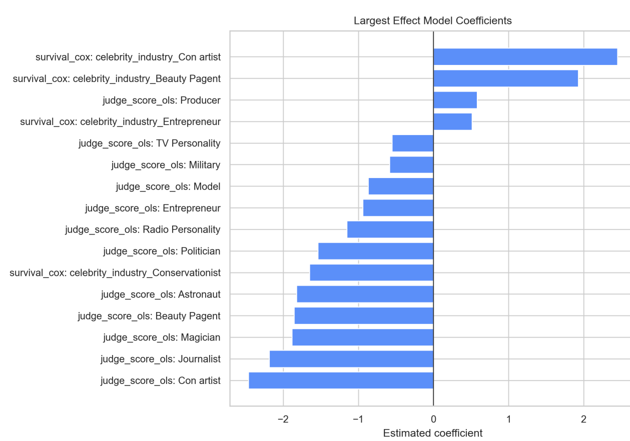


Figure 13: Exploratory effect coefficients with uncertainty intervals.

10 Proposed Voting System

The central design lesson is that close bottom-two cases should be treated as ambiguous rather than forced into a deterministic public-facing ranking. We therefore propose a hybrid system:

$$\text{Percent Rule + Gray-Zone Judge Save.}$$

The system first converts judge scores and fan votes into percentages and adds them. If the combined-score gap between the lowest two contestants is larger than a threshold τ, the lowest combined score is eliminated. If the margin is at or below τ, the bottom two enter a judge-save decision. In our simulation, τ is set to 0.00354, the 25th percentile of positive modeled bottom-two combined-score margins. For quantile level q, the general threshold is

$$\tau_q = Q_q\{C^{\text{pct}}_{(2),t} - C^{\text{pct}}_{(1),t} : C^{\text{pct}}_{(2),t} > C^{\text{pct}}_{(1),t}\},$$

where $C^{\text{pct}}_{(1),t}$ and $C^{\text{pct}}_{(2),t}$ are the lowest and second-lowest combined percent scores in week t. The proposed decision rule is

$$\widehat{E}_t = \begin{cases} \arg\min_i C^{\text{pct}}_{i,t}, & C^{\text{pct}}_{(2),t} - C^{\text{pct}}_{(1),t} > \tau_q, \\ \arg\min_{i \in B_t} J_{i,t}, & C^{\text{pct}}_{(2),t} - C^{\text{pct}}_{(1),t} \le \tau_q, \end{cases}$$

where B_t is the bottom-two set. The second line approximates a judge save by keeping the stronger judge-scored contestant when the combined margin is too small to treat as decisive.

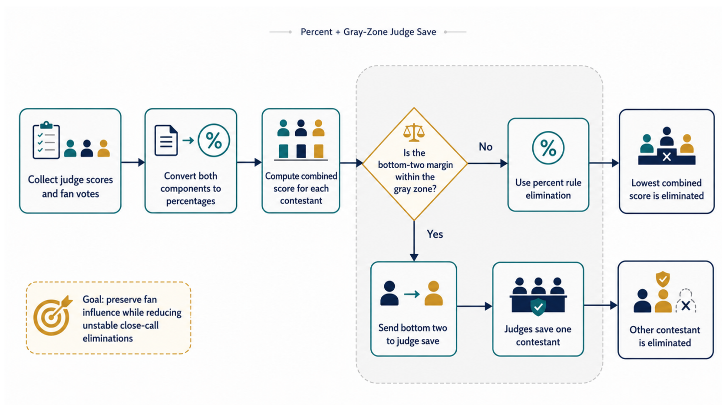


Figure 14: Proposed Percent + Gray-Zone Judge Save workflow.

The proposed system has three advantages. First, percent scoring is more transparent than rank aggregation because judge and fan shares are measured on the same scale. Second, the gray zone acknowledges that tiny margins are not substantively robust when fan support is hidden and highly non-identifiable. Third, judge discretion is limited to close bottom-two cases, which preserves fan influence while giving producers a defensible way to manage fragile eliminations.

Table 21 compares the main rule choices against the criteria introduced at the start of the paper. A pure rank rule is easy to explain but discards score magnitudes. A pure percent rule is transparent and audience-centered, but it still treats tiny hidden-vote margins as decisive. The proposed hybrid keeps the percent scale while adding an explicit safeguard only where the modeled decision is fragile.

Table 21: Rule-design scorecard.

Criterion	Rank rule	Percent rule	Bottom-two save	Proposed hybrid
Uses score magnitude	Weak	Strong	Depends on trigger	Strong
Protects fan influence	Medium	Strong	Medium	Strong outside gray zone
Handles close calls	Weak	Weak	Strong	Strong
Viewer transparency	Medium	Strong	Medium	Strong with audit label
Reproducible audit path	Medium	Strong	Weak if discretionary	Strong

Table 22: Operational protocol for the proposed voting system.

Stage	Rule action	Audit output
Score normalization	Convert judges' totals and fan votes to percentage shares.	Published judge share and fan share after the voting window.
Combined ranking	Compute $C_i^{\text{pct}} = P_i^J + F_i$.	Public bottom two and combined percent margin.
Gray-zone test	Compare the bottom-two gap with τ_q.	"Clear decision" or "close-call judge save" label.
Judge save	If triggered, judges save one contestant from the bottom two.	Recorded judge-save decision and reason.
Post-show audit	Archive scores, threshold, and decision path.	A reproducible elimination record for producers.

Table 23: Producer-facing risks and controls in the proposed rule.

Risk	Why it matters	Control in the proposed system
Hidden-vote opacity	Viewers cannot verify exact public vote totals.	Publish judge share, fan share, combined share, and bottom-two margin after the voting window.
Close-call backlash	Very small margins can look arbitrary.	Label close cases as gray-zone decisions and use the judge save only there.
Rank compression	Rank aggregation can erase large score gaps.	Use percent shares so score magnitudes remain visible.
Excess judge control	A broad save power could weaken the audience vote.	Restrict judge discretion to the bottom two and to margins below τ.
Audit failure	Post-show explanations may appear improvised.	Archive the decision path, threshold, and score table for every modeled week.

Table 24: Simulation summary for the proposed gray-zone system.

Metric	Value
Gray-zone threshold τ	0.00354
Gray-zone trigger rate	61.7%
Proposed system differs from pure percent rule	46.0%
Proposed elimination appears in observed set	54.0%

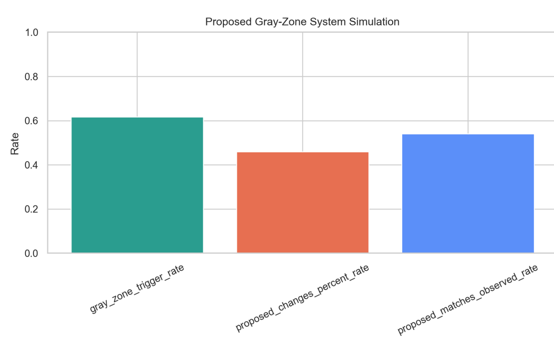


Figure 15: Simulation dashboard for the proposed system.

11 Sensitivity Analysis

The proposed system depends on the gray-zone threshold τ, so we test how the recommendation changes when τ is chosen from different quantiles of the positive bottom-two margin distribution. The "No gray zone" row in Table 25 is the pure percent reconstruction. It matches the historical modeled eliminations by construction, but it offers no special treatment for statistically fragile close calls.

Table 25: Sensitivity of the proposed system to the gray-zone threshold.

Threshold	Trigger rate	Changes percent	Historical match
No gray zone	0.000	0.000	1.000
5th pct.	0.513	0.460	0.540
10th pct.	0.540	0.460	0.540
25th pct.	0.617	0.460	0.540
50th pct.	0.743	0.460	0.540
75th pct.	0.870	0.460	0.540
90th pct.	0.946	0.460	0.540

The sensitivity curve has a useful interpretation. Once the threshold is large enough to activate the judge save in the genuinely close cases, the set of weeks where the proposed system differs from pure percent stabilizes at 46.0%. Increasing τ further sends more weeks to the gray-zone procedure, but many of those additional weeks still eliminate the same contestant because the judge-weaker bottom-two contestant is already the percent-rule loser. We use the 25th percentile threshold as a middle setting: it captures close calls without turning the judge save into the default elimination mechanism.

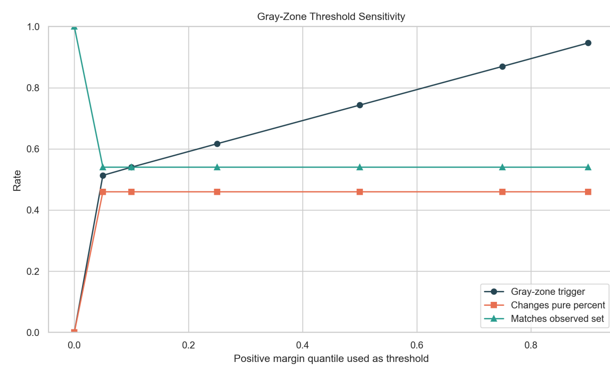


Figure 16: Gray-zone threshold sensitivity.

11.1 Robustness of the main conclusions

The major conclusions do not depend on a single threshold value. First, hidden fan support remains highly non-identifiable across all settings; the mean feasible fan-share interval width is 0.862 before the gray-zone rule is even introduced. Second, rank and percent rules disagree in 26.4% of modeled weeks under the same fan-support estimates, so rule choice is substantively important. Third, the judge-only baseline remains weak, matching only 36.4% of historical eliminated sets. The threshold mainly controls how much explicit judge discretion producers want in close calls; it does not change the underlying evidence that close-call transparency is needed.

12 Quality Control and Reproducibility

We separated reproducibility checks from substantive interpretation. A model can be reproducible but still answer the wrong question, so every major output was tied to a specific check: data reconstruction, constraint feasibility, baseline comparison, rule counterfactuals, and recommendation stability. This mirrors the practical workflow a producer would need if the rule were audited after a controversial elimination.

Table 26: Quality-control checks used before drawing conclusions.

Check	Evidence in this paper	Reason for inclusion
Data reconstruction	2,777 active contestant-week records and 261 modeled elimination weeks.	Confirms that the weekly competitive panel is usable.
Constraint feasibility	Historical modeled eliminations are feasible under the percent-rule inequalities.	Confirms that the latent fan-support problem is internally consistent.
Baseline benchmark	Judge-only baseline matches 36.4% of historical eliminated sets.	Shows that fan support is necessary, not decorative.
Rule counterfactual	Rank and percent rules differ in 26.4% of weeks.	Demonstrates that aggregation choice changes outcomes.
Uncertainty audit	Mean feasible fan-share width is 0.862.	Prevents overclaiming exact public-vote recovery.
Recommendation stability	Threshold sensitivity preserves the main conclusion across tested quantiles.	Shows that the gray-zone idea is not a single-parameter artifact.

All reported figures and tables are generated from the same processed data pipeline. The paper therefore treats visualizations as audit artifacts rather than illustrations added after the fact. If a number appears in the memo or recommendation, it is traceable to one of the diagnostics above.

13 Viewer Communication and Audit Package

A voting rule also needs a communication rule. The show should not reveal raw vote counts if the network treats them as confidential, but it can still reveal the decision path in a reproducible way. We propose that each elimination be stored as an audit tuple

$$\mathcal{A}_t = \{P_i^J,\ F_i,\ C_i^{\text{pct}},\ C_{(2),t}^{\text{pct}} - C_{(1),t}^{\text{pct}},\ \tau_q,\ L_t\},$$

where L_t is the public decision label: clear percent decision or gray-zone judge save. This tuple is enough to explain the result without exposing raw ballot totals. It also makes the proposed system harder to criticize as an after-the-fact producer decision because the threshold and label are computed before the judge-save choice is announced.

The communication rule should be conservative. If the bottom-two margin is larger than τ_q, producers should say that the combined percentage result was clear. If the margin is no larger than τ_q, producers should say that the result entered the gray zone and that judges made the limited bottom-two save. The show should not claim that a contestant had an exact level of public support unless the raw vote share is actually released. This distinction is important because our model shows that hidden fan support is highly non-identifiable. Clear language prevents the same mistake a weak model would make: treating an estimated or confidential fan component as if it were a publicly verified exact number.

This audit package also improves the fairness of controversial cases. In a fan-rescue case, the public can see that fan support changed the judge-only bottom set. In a judge-favored risk case, the public can see that high judge scores did not guarantee safety once fan share was included. In a gray-zone case, the public can see that the judges intervened only because the combined margin was small. Thus the same reporting structure explains the three most important kinds of viewer complaints.

Table 27: Recommended public audit package after each elimination.

Released item	Calculation	Purpose
Judge percentage	$P_i^J = J_i / \sum_k J_k$.	Shows how expert scoring contributes on the same scale as fan voting.
Fan percentage rank	Rank induced by verified fan share F_i.	Confirms that public voting affected the combined order.
Combined percentage	$C_i^{\text{pct}} = P_i^J + F_i$.	Gives viewers the direct rule used for the bottom-two comparison.
Bottom-two margin	$C_{(2),t}^{\text{pct}} - C_{(1),t}^{\text{pct}}$.	Distinguishes clear eliminations from fragile close calls.
Decision label	Clear decision if margin $> \tau_q$; gray-zone judge save otherwise.	Makes judge discretion conditional rather than open-ended.
Archived path	Scores, threshold, label, and save decision.	Allows a later audit to reproduce the public explanation.

14 Memo to DWTS Producers

To: DWTS Producers

Subject: A transparent voting system for close eliminations

The analysis suggests that fan votes should remain central to the show, but close bottom-two cases should not be treated as if the hidden fan vote were fully observable. We recommend replacing opaque rank aggregation with a Percent + Gray-Zone Judge Save system.

- Convert judge scores and fan votes into percentages before combining them. This makes the rule easier to explain to viewers.
- Use the standard percent result when the bottom-two margin is clearly separated.
- When the bottom-two margin is inside the gray zone, ask judges to save one contestant from the bottom two.
- Publicly describe the judge-save condition as a close-call safeguard, not as a replacement for fan influence.

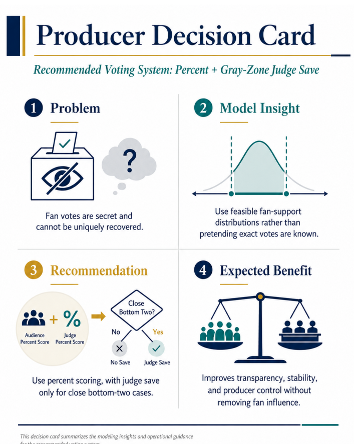


Figure 17: Producer decision card.

15 Strengths, Limitations, and Extensions

Strengths. The model respects the fact that fan votes are hidden. It does not invent labels or claim to recover exact vote totals. Every numerical result in the paper is generated from the same reproducible workflow, and the validation step confirms that the reconstructed constraints are internally consistent with the observed modeled eliminations.

Limitations. The maximum-entropy fan profile is a conservative point estimate, not a direct observation. The bottom-two judge-save simulation is a policy model because detailed historical judge-save ballots are not fully available in the data. The professional-dancer and celebrity-effect regressions are observational associations and may reflect unmodeled casting, editing, or audience factors.

Extensions. Future work could add verified external popularity signals, such as social media engagement or search interest, but only if those sources are logged and validated. Another extension would model the evolution of fan support across weeks with a state-space model. We did not use those extensions in the main workflow because the official data already support a complete and transparent constraint-based solution.

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